EU CBAM Embedded Emissions Verification Certificate

Prepared in accordance with European Union Regulation (EU) 2023/956 & Commission Implementing Regulation (EU) 2023/1773 for Non-EU MSME Installations.

1. Installation, Product & Consignment Scope

Exporter / Installation	Apex Forgings & Steel Products Pvt Ltd	**Manufacturing Facility**	Unit 3 Induction Furnace & Rolling Mill (Ludhiana, Punjab, India)
Export Product Name	Hot-Rolled Structural Alloy Steel Bars (Fe500D)	**Product Category**	Iron and Steel
Consignment Quantity	**48.66 MT**	**CBAM CN Code**	7214 20 00
Destination Market	Germany	**Reporting Period**	Q1 2026 (2026)
Upstream Supplier	Deccan Sponge Iron & Power Ltd	**Authorized EU Buyer**	Krupp Bautechnik GmbH

2. Certified Embedded Emissions Summary

Emissions Source Scope	Activity Data	Applied Emission Factor	Embedded (tCO2e)
Scope 1: Direct Combustion Stationary furnace fuels & burners	1727.4 Liters	3.114 kg CO2/L (IPCC)	**5.3800**
Scope 2: Indirect Electricity Grid consumption for production line	20437.2 kWh	0.716 kg CO2/kWh (CEA India)	**14.6300**
Scope 3: Precursor Materials Upstream embedded raw materials (DRI / Pig Iron)	52.55 MT	1.92 tCO2/MT (EU Default)	**100.9000**
TOTAL EMBEDDED EMISSIONS	—	Annex IV Summation	**120.9100 tCO2e**
SPECIFIC EMBEDDED EMISSIONS INTENSITY	Consignment: 48.66 MT	Formula: Total / Qty	**2.4848 tCO2e / MT**

3. Multi-Agent Source Document Evidence & Traceability Trail

Activity Field	Source Document	Extracted Value	Confidence
Export Invoice Net Quantity	Commercial_Export_Invoice_APEX_2026_044.pdf Page 1 | Body text	**48.66 MT**	99%

Mill Test Certificate Certified Net Mass	Mill_Test_Certificate_HT88421.pdf Page 1 | Body text	**48.66 MT**	99%
PSPCL Industrial Electricity Reading	PSPCL_Industrial_Electricity_Bill_Q1.pdf Page 1 | Body text	**20,437.2 kWh**	98%
Furnace Oil Fuel Consumption	IOCL_Furnace_Oil_Fuel_Invoice_981.pdf Page 1 | Body text	**1,727.4 Liters**	97%
Supplier DRI Raw Material Precursor Lot	Deccan_Sponge_DRI_Emissions_Cert.pdf Page 1 | Body text	**52.55 MT Sponge Iron**	96%

4. Verifier Agent Automated Audit Results

Audit Check	Severity	Result	Audit Finding Description
Multi-Document Quantity Reconciliation	INFO	**PASSED**	Verified: Product quantity (48.660 MT) matches across invoice and test certificate.
Summation Integrity Verification	INFO	**PASSED**	Arithmetic confirmed: 5.38 + 14.63 + 100.90 = 120.91 tCO2e.
Specific Emissions Intensity Verification	INFO	**PASSED**	Intensity mathematically confirmed at 2.4848 tCO2e / MT.
Grid Emission Factor Applicability	INFO	**PASSED**	CEA Baseline factor (0.716 kg CO2/kWh) compliant with Indian benchmarks.

Methodology & Regulatory Notice: This audit report was compiled by CarbonGuard AI's multi-agent compliance engine using deterministic arithmetic. Calculations adhere to European Union CBAM transitional and definitive rules. Central Electricity Authority baseline factors were applied for Scope 2 calculations. Automated internal checks do not replace accredited third-party verifier certification where explicitly mandated by national authorities.